

EMILY LIANG

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Education

The University of Texas at Austin

May 2028 (Expected)

*B.S. Computer Science, **Turing Scholars Honors**; B.S. Mathematics; Minor in Business*

Austin, TX

Relevant Coursework

Programming Languages, Artificial Intelligence, Algorithms, Machine Learning, Operating Systems, Computer Architecture, Compilers, Computer Vision, Probability, Stochastic Processes, Number Theory

Experience

Quantitative Developer Intern

May 2026 – Aug. 2026

Base Power Company

Austin, TX

- Developed an optimization-based battery dispatch algorithm under ERCOT market and operational constraints, improving expected profitability by approximately \$2/kWh.
- Built simulation and cross-validation infrastructure for battery dispatch strategies across historical ERCOT market conditions, reducing backtest error by 15%.
- Designed statistical experiments to quantify strategy performance, analyze sensitivity to market conditions, and guide optimization decisions.

Undergraduate Course Assistant

Aug. 2026 – Present

The University of Texas at Austin

Austin, TX

- Mentor students in Software Engineering and Freshman Think Lab on software architecture, debugging, testing, version control, and project development.
- Lead office hours and conduct code and design reviews, providing individualized technical feedback on implementation and software engineering practices.

Simulation and Validation System Lead

Aug. 2024 – Present

Longhorn Racing Internal Combustion

Austin, TX

- Lead a team of 7 engineers responsible for simulation, telemetry, and validation infrastructure for a Formula SAE race car, setting technical priorities and coordinating integration across vehicle systems.
- Direct simulation and track-validation efforts across vehicle dynamics, testing, driver development, and race strategy to support engineering design decisions.
- Coordinate validation plans with subsystem leads to translate vehicle data and simulation results into model calibration and design recommendations.

Software Engineering Intern

Sept. 2023 – May 2024

Lockheed Martin Aeronautics

Fort Worth, TX

- Collaborated with the Air Force Research Laboratory, L3Harris, and Raytheon on research supporting satellite communications capabilities for the F-35 platform.

Projects

Longhorn Racing Lap-Time Simulation | *Python, MATLAB*

- Developed a quasi-steady-state lap-time simulator modeling aerodynamic, vehicle dynamics, and powertrain constraints to predict Formula SAE vehicle performance across race tracks.
- Integrated GGV performance envelopes with track trajectories and configurable vehicle models to compute feasible velocity profiles and predicted lap times.
- Implemented parameter sweeps to quantify lap-time sensitivity to vehicle design variables, enabling simulation-driven comparison of engineering configurations.

JPEB: Custom 16-Bit Computer | *Rust, Haskell, C, Verilog, Python, Assembly*

- Designed and implemented a custom 16-bit computer in a team of four, spanning ISA design, processor architecture, compiler tooling, and emulation.
- Developed a 6-stage pipelined processor in Verilog and deployed it to an FPGA, implementing hazard detection, data forwarding, pipeline stalls, and control-flow flushing.
- Extended the compiler and assembler toolchain and implemented Chrome Dinosaur and Snake using the custom ISA to validate end-to-end hardware and software execution.

x86 Multicore Operating System Kernel | *C++*

- Built a preemptive multicore x86 operating system kernel supporting concurrent process and thread execution with synchronization primitives.
- Implemented virtual memory with demand paging, page caching, swapping, and memory-mapped files.
- Developed kernel-level process, thread, file-system, and memory-management system calls.

Technical Skills

Languages: C/C++, Python, Go, Java, JavaScript/TypeScript, Verilog, x86 Assembly, ARM Assembly, OCaml

Technologies: Git, Docker, Linux, NumPy, SciPy, Pandas, Matplotlib, PyTorch